TinyML Milestone 2

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1 (R1) Dataset Description

1.1 Class definitions (classifications)

Table 1: Activity class definitions for UCI-HAR and WISDM datasets

Class name	Class definition
WALKING (UCI-HAR)	Subject performs continuous forward movement on flat ground with alternating steps at a natural pace and no elevation change.
WALKING UPSTAIRS (UCI-HAR)	Subject ascends a staircase by stepping upward one or more steps at a time, increasing elevation with each step.
WALKING DOWNSTAIRS (UCI-HAR)	Subject descends a staircase by stepping downward one or more steps at a time, decreasing elevation with each step.
SITTING (UCI-HAR)	Subject remains stationary in a seated posture on a chair with minimal movement.
STANDING (UCI-HAR)	Subject remains stationary in an upright posture on both feet with minimal movement.
LAYING (UCI-HAR)	Subject lies horizontally on a surface with the torso parallel to the ground and no upright support.
Walking (WISDM)	Subject performs continuous forward movement on flat ground with alternating steps at a natural pace and no elevation change.
Jogging (WISDM)	Subject performs forward movement at a faster pace than walking, including brief periods where both feet are off the ground.
Upstairs (WISDM)	Subject ascends a staircase by stepping upward one or more steps at a time, increasing elevation with each step.
Downstairs (WISDM)	Subject descends a staircase by stepping downward one or more steps at a time, decreasing elevation with each step.
Sitting (WISDM)	Subject remains stationary in a seated posture on a chair with minimal movement.
Standing (WISDM)	Subject remains stationary in an upright posture on both feet with minimal movement.

1.2 Sample counts per class:

Table 2: Class distribution table for UCI-HAR dataset.

Class	**Train**	**Val**	**Test**	**Total**
WALKING	888	338	496	1722
WALKING_UPSTAIRS	797	276	471	1544
WALKING_DOWNSTAIRS	744	242	420	1406
SITTING	993	293	491	1777
STANDING	1053	321	532	1906
LAYING	1076	331	537	1944
Total	5551	1801	2947	10299

1.3 Additional requirements for Track B (open/public dataset):

Dataset identity: Two publicly available datasets were used to train the baseline model prior to edge deployment: the Human Activity Recognition Using Smartphones (UCI HAR) dataset v1.0 and the Wireless Sensor Data Mining (WISDM) dataset v1.1, available at https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using and https://www.cis.fordham.edu/wisdm/dataset.php, respectively. The UCI HAR dataset is released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license, while the WISDM dataset does not specify a formal license and instead requests citation of the original work. Both datasets are publicly accessible and widely used for academic research.

Why this dataset: Both datasets are widely recognized as benchmarks in human activity recognition (HAR) research and provide large-scale, well-labeled data from multiple subjects. They include common activities of daily living (ADL) which support robust multi-class classification and align with our goal of ADL recognition. Their widespread use enables fair comparison with state-of-the-art (SOTA) methods, while their diversity supports better generalization and robustness. For this milestone 2, UCI HAR is selected as the primary dataset due to its close alignment with the target deployment setting, including a 50 Hz sampling rate, combined accelerometer and gyroscope sensors, and waist-mounted sensor placement. In contrast, WISDM is used as a secondary dataset for domain-gap analysis, particularly for differences in sensor placement and modality, and will support future accelerometer-only ablation studies in Milestones 3–4.

Dataset characteristics: The UCI HAR dataset was collected from 30 volunteers performing six activities while wearing a Samsung Galaxy S II smartphone at the waist and recorded 3-axis linear acceleration and angular velocity at a sampling rate of 50 Hz with built-in accelerometer and gyroscope. The data was preprocessed by applying noise filtering and segmented into 2.56-second sliding windows with 50% overlap that resulted in 10,299 pre-windowed samples. Each window is represented by a 561-dimensional feature vector derived from time- and frequency-domain statistics for activity classification. The WISDM dataset contains accelerometer-only data collected from 36 users performing six activities (Walking, Jogging, Upstairs, Downstairs, Sitting, and Standing). Data was recorded using a smartphone at a sampling rate of 20 Hz, producing 1,098,207 raw time-series samples. Each record includes 3-axis acceleration (x, y, z), along with user ID, activity label, and timestamp. A transformed version is also provided, where 10-second segments (200 samples) are converted into 46-dimensional feature vectors using statistical and peak-based features, resulting in 5,424 preprocessed samples.

Domain gap acknowledgment: A domain gap exists between the public datasets (UCI HAR and WISDM) and the target deployment using the Arduino Nano 33 BLE Sense with its onboard LSM9DS1 IMU. The primary difference lies in sensor hardware, as both datasets use smartphone-based sensors, whereas the deployment relies on a dedicated embedded IMU with different calibration, sensitivity, noise characteristics, and signal quality. In terms of sensor modality, sampling rate, and placement, UCI HAR is closely aligned with the deployment, as it includes both accelerometer and gyroscope data, uses a 50 Hz sampling rate, and employs waist-mounted placement. In contrast, WISDM exhibits a larger domain gap, as it provides accelerometer-only data, operates at 20 Hz, and uses thigh-based placement. Additionally, WISDM does not include the LAYING class required for this task. These differences lead to a shift in data distribution where patterns learned during training may not fully generalize to the deployment environment. Consequently, on-device performance may degrade due to differences in sensor characteristics, preprocessing, and real-world user behavior.

Arduino data collection plan: For Milestone 3, we will collect an Arduino-based inertial dataset using the Arduino Nano 33 BLE Sense with its onboard LSM9DS1 IMU (accelerometer and gyroscope). Data will be recorded at 50 Hz with waist-mounted placement and segmented into 2.56-second windows with 50% overlap. The activity

classes include WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, and LAYING. Data will be collected from 1–2 users across at least two indoor environments and multiple sessions on different days (e.g. 22-23th April), with a minimum of 100 labeled windows per class after segmentation.

2 (R2) Data Card

Table 3: Required Data Card fields.

Field	**Your Entry**	**Track**
Task type	Multi-class classification (human activity recognition)	B
Number of classes (or anomaly definition)	UCI HAR: 6 (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING) WISDM: 6 (Walking, Jogging, Upstairs, Downstairs, Sitting, Standing)	B
Sensor(s) and sampling rate	UCI HAR: accelerometer and gyroscope; 50 Hz sampling rate WISDM: accelerometer only; 20 Hz sampling rate	B
Window length and overlap	UCI HAR: 2.56 s windows with 50% overlap WISDM: 10 s windows with no overlap	B
Features extracted (if any)	UCI HAR: 561 features per window from acc and gyro; time- and frequency-domain features (XYZ + magnitude), including statistical, FFT-based, and angle features WISDM: 43 features per window from acc; time-domain (XYZ + magnitude) with statistical and peak-based features	B
Total samples	UCI HAR: 10,299 windowed samples WISDM: 5,424 windowed samples	B
Train / Val / Test split ratio	UCI HAR: 70% / 30% (train/test; no validation set) WISDM: not predefined (no official split)	B
Split method	UCI HAR: per-subject WISDM: not predefined (user attribute available; split can be user-defined)	B
Number of subjects/users	UCI HAR: 30 users WISDM: 36 users	B
Number of environments/locations	Not collected yet (planned to be 1-2 indoor environments)	A
Collection dates	Not collected yet (planned to be 22-23th April)	A
Known limitations	UCI HAR is smartphone waist-mounted and preprocessed; WISDM lacks LAYING and gyroscope data; both differ from LSM9DS1 IMU.	B
Track B additional fields:		
Dataset name and URL/citation	UCI HAR v1.0: `https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using` WISDM v1.1: `https://www.cis.fordham.edu/wisdm/dataset.php`	B
License	UCI HAR: CC BY 4.0 WISDM: not explicitly licensed (citation requested)	B
Original recording equipment	UCI HAR: Samsung Galaxy S II (accelerometer + gyroscope) WISDM: smartphone accelerometer (Android; model not specified)	B
Domain gap vs. Arduino	UCI HAR: Galaxy S II IMU vs LSM9DS1 IMU; differences in sensor sensitivity, noise characteristics, and calibration. WISDM: smartphone accelerometers vs LSM9DS1 IMU; differences in sensor sensitivity, noise characteristics, calibration, sampling (20 Hz vs 50 Hz), placement (thigh vs waist), and modality (acc only vs acc + gyro)	A & B
Arduino collection plan for M3	Collect LSM9DS1 IMU acc+gyro data at 50 Hz (waist-mounted); 2.56 s windows, 50% overlap; 1–2 users, 2 indoor environments, 2 sessions (different days); $\geq$100 labelled windows per class	A

3 (R3) Preprocessing Pipeline

1. Segmentation: UCI HAR uses fixed sliding windows provided by the dataset with 50 Hz sampling, 128 samples per window (2.56 s), and 50% overlap (64-sample stride). Each sample is already pre-segmented, so no additional windowing is applied. Input tensors have shape $[128, 6]$ using total acceleration (x/y/z) and body gyroscope (x/y/z). The windows are generated using fixed periodic sliding, not event-triggered segmentation.

2. Feature extraction: No handcrafted time-domain or frequency-domain features are computed. The model is trained directly on normalized raw inertial signal windows using 3-axis accelerometer and gyroscope data ($[128, 6]$). The UCI HAR engineered 561-feature vectors are not used to avoid computational overhead for on-device deployment.

3. Normalization: Per-channel normalization is applied to each of the six sensor channels (total acceleration and body gyroscope x/y/z). First, standardization is performed using mean and standard deviation computed from the training set only. The standardized values are then scaled using min–max normalization. All normalization parameters are computed from the training set and applied unchanged to validation and test sets to prevent information leakage.

4. Data augmentation: No data augmentation is used in the reportable Milestone 2 runs because the goal at this stage is to establish a clean and controlled benchmark.

5. Additional Info: The UCI HAR dataset provides an official subject-disjoint train/test split with 21 subjects for training and 9 unseen subjects for testing. The test set is kept untouched throughout preprocessing and training to prevent data leakage. A validation set is created from the training subset using a 75%/25% split with subject-aware grouping. Labels are converted from 1–6 to zero-based class indices (0–5) for compatibility with Keras/TensorFlow.

For WISDM, the raw semicolon-delimited rows are parsed to extract subject, activity, timestamp, and $x/y/z$ acceleration. The acceleration scale is converted using the dataset note that 10 corresponds to $1g$. Timestamp inspection showed irregular effective sampling, so WISDM is used as secondary dataset for robustness evidence and domain-gap analysis rather than as the main six-class benchmark (therefore our models have not trained on this dataset as of Milestone 2). Jogging is not forced into the UCI label set, and LAYING is not artificially introduced.

Overall, the preprocessing pipeline prioritizes reproducibility, subject-disjoint evaluation, and consistency between the heavy offline baseline and the deployable lightweight model.

Table 4: Preprocessing and data handling decisions in Milestone 2

Item	**UCI HAR setup**	**WISDM setup**	**Milestone 2 decision**
Input format	Pre-windowed inertial sequences	Parsed raw accelerometer rows	Keep raw sequential input for neural models
Channels used	Total acc $x/y/z$ + body gyro $x/y/z$	Acc $x/y/z$ only	Do not force incompatible sensor merging
Split method	Official subject-disjoint test split; validation from training subjects	No official split	Use UCI HAR as main benchmark
Normalization	Computed on training only	Applied after parsing / optional segmentation	Reuse unchanged statistics on val/test
Sampling	50 Hz fixed windows	Nominal 20 Hz with irregular timestamps	Audit timestamps; use as secondary evidence
Augmentation	None	None in reportable run	Keep evaluation controlled and reproducible

4 (R4) Baseline Model

Two model paths were evaluated in this milestone under the same standardized preprocessing pipeline.

The first path is a paper-reproduction baseline [1], implemented as an offline TensorFlow/Keras approximation of the reference stacking framework. It combines five level-0 hybrid deep learning models — ConvLSTM, CNN-GRU, CNN-BiGRU, CNN-BiLSTM, and CNN-LSTM — with an XGBoost meta-learner at level 1. Its role is to serve as a strong offline reference rather than a deployable model.

The second path is the lightweight TinyML-oriented proposal. This model is a depthwise-separable 1D CNN with only 1,922 Keras parameters. It uses the same $[128, 6]$ input format and predicts the same six activity classes, but is designed to be compact enough for later TensorFlow Lite conversion, quantization, and microcontroller deployment.

Both models are trained using Adam with learning rate 0.001 and sparse categorical cross-entropy loss. The paper-reproduction baseline is run in a bounded 20-epoch configuration, while the lightweight model is trained for up to 40 epochs with patience 8.

Table 5: Baseline and improved model paths

Model path	Architecture	Input / output	Training setup	Deployment intent
Paper reproduction baseline	ConvLSTM, CNN-GRU, CNN-BiGRU, CNN-BiLSTM, CNN-LSTM + XGBoost stack	$[128, 6] \rightarrow$ 6-class softmax	Adam lr 0.001, decay 0.01, batch 50, 20-epoch bounded run	Offline reference only
Lightweight proposal	Depthwise-separable 1D CNN (1,922 parameters)	$[128, 6] \rightarrow$ 6-class softmax	Adam lr 0.001, decay 0.01, batch 50, 40 epochs, patience 8	Primary TinyML candidate

Four-Phase Development Roadmap

Table 6: Four-phase development roadmap

Phase	Title	Objective	Main change	Evaluation focus
1	Rebuilding the paper baseline	Reproduce the reference HGAR pipeline as closely as feasible to establish a strong upper-bound baseline.	No intentional simplification.	Accuracy
2	Lightweight model screening	Compare TinyML-friendly models under the same data pipeline and select the best deployable candidate.	Architecture only; still in progress.	Accuracy + latency
3	Eco-mode sensor ablation	Test whether accelerometer-only sensing is worthwhile after the lightweight candidate is chosen.	Sensor configuration only; remove gyroscope for comparison.	Energy + accuracy
4	Quantized edge deployment	Deploy the selected final model on Arduino using TFLM and evaluate real edge behavior.	INT8 quantization and on-device implementation.	Energy + accuracy + latency

5 (R5) Classification Baseline Results (Held-Out Test Set)

This section presents the classification performance of the baseline model and the lightweight 1D CNN model, evaluated on the per-subject test subset of the UCI HAR dataset. The test set consists of 2,947 windows from 9 unseen users. The evaluation includes overall accuracy, precision, recall, and F1-score, as well as per-class metrics with the number of test samples per class, as summarized in Table 7. Additionally, confusion matrices for both models are shown in Figures 1 and 2.

Table 7: Per-class performance comparison of Baseline and TinyCNN models.

	Baseline (Acc = 0.914)				1D CNN (Acc = 0.917)			
Class	Prec	Rec	F1	Sup	Prec	Rec	F1	Sup
WALKING	0.9637	0.9093	0.9357	496	0.9857	0.9718	0.9787	496
WALKING_UPSTAIRS	0.9447	0.9427	0.9437	471	0.9571	0.9469	0.9520	471
WALKING_DOWNSTAIRS	0.8460	0.9548	0.8971	420	0.9213	0.9476	0.9343	420
SITTING	0.8091	0.8717	0.8392	491	0.8273	0.7902	0.8083	491
STANDING	0.9190	0.8102	0.8611	532	0.8141	0.8477	0.8306	532
LAYING	1.0000	1.0000	1.0000	537	1.0000	1.0000	1.0000	537
Macro avg	0.9137	0.9148	0.9128	2947	0.9176	0.9174	0.9173	2947
Weighted avg	0.9167	0.9135	0.9137	2947	0.9172	0.9169	0.9168	2947

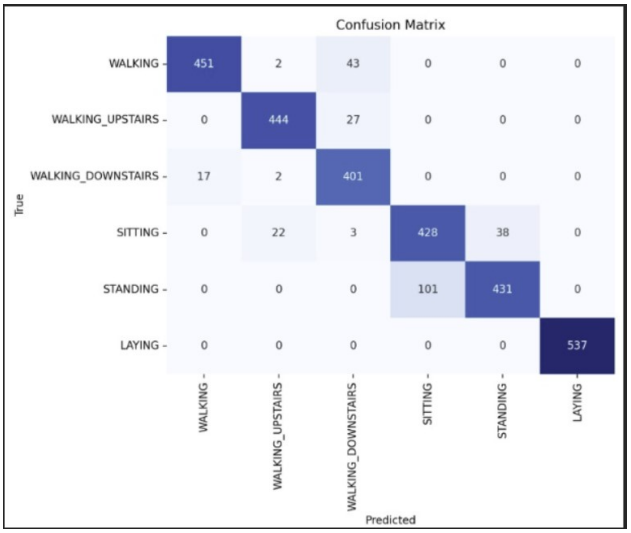

Figure 1: Confusion matrix (baseline)

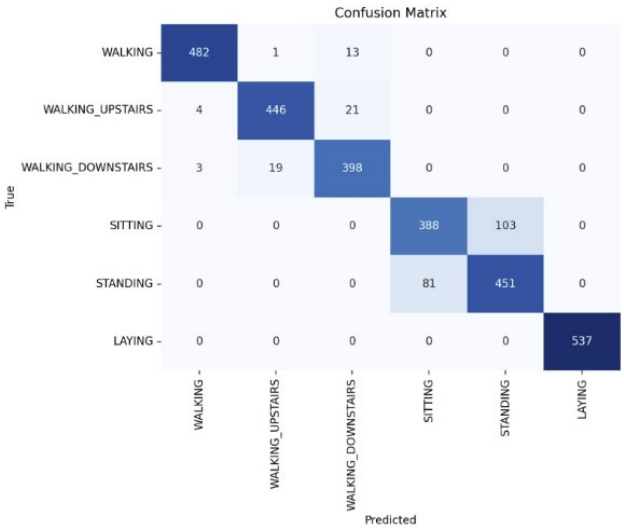

Figure 2: Confusion matrix (lightweight)

6 (R6) Error Analysis

Top confusion pairs: The dominant errors in both models occur between the static classes SITTING and STANDING. In the baseline model, SITTING is misclassified as STANDING 103 times, while STANDING is misclassified as SITTING 81 times. A similar pattern is observed in the lightweight model, with 38 misclassifications from SITTING to STANDING and 101 in the reverse direction. Whereas LAYING is perfectly classified due to its distinct static orientation and minimal overlap with other activities.

Hypothesized causes: SITTING and STANDING are static activities with minimal motion and similar posture, making them difficult to distinguish using gravity and orientation cues alone. Similar overlap in sensor signatures is also observed among walking-related classes (e.g., WALKING_UPSTAIRS and WALKING_DOWNSTAIRS), which share comparable periodic motion patterns.

Concrete next steps: First, explore lightweight feature extraction and feature selection to identify a small set of discriminative features for separating SITTING and STANDING particularly static posture separation with minimal additional overhead for on-device use. Second, evaluate alternative loss functions, such as focal loss, in Milestone 3 to place greater training emphasis on hard classes (e.g., SITTING and STANDING) rather than already easy classes such as LAYING.

Domain gap analysis: Although UCI HAR is closely aligned with the target deployment, a domain gap remains due to differences in sensor hardware. The dataset was collected using a Samsung Galaxy S II smartphone, whereas deployment uses the Arduino Nano 33 BLE Sense with an onboard LSM9DS1 IMU. This leads to differences in calibration, sensitivity, noise characteristics, and overall signal quality.

Sampling rate (50 Hz), sensor modality (accelerometer + gyroscope), and waist-mounted placement are matched to minimize the gap. However, real-world deployment may still differ due to environmental conditions and user variability,

introducing shifts in motion patterns and noise levels.

These differences may degrade performance, particularly for subtle classes such as SITTING and STANDING, which rely on small orientation and gravity cues. To mitigate this, the deployment pipeline will follow the same preprocessing as UCI HAR (2.56 s windows with 50% overlap) and reuse training-based normalization. In addition, lightweight feature extraction and calibration, along with custom Arduino data collection, will be used to better align real sensor data with the training distribution.

7 (R7) Updated Plan for M3

Milestone 3 will continue Phases 2, 3, and 4 rather than stopping at the first lightweight result. Phase 2 remains active because we are still exploring additional lightweight architectures under the same standardized preprocessing pipeline. Phase 3 will explicitly test an accelerometer-only configuration without gyroscope data to determine whether sensing cost can be reduced while maintaining acceptable performance. Phase 4 will quantize the selected final model and move it toward TensorFlow Lite Micro deployment on the Arduino Nano 33 BLE Sense.

In addition to architecture screening and deployment, Milestone 3 will also address the public-data to embedded-device domain gap. This will be done by collecting Arduino-based hold-out data, comparing public benchmark results against embedded-device evaluation, and reporting public-dataset performance separately from Arduino deployment performance.

Table 8: Updated plan for Milestone 3

Item	**Planned M3 action**	**Target / condition**	**Why it matters**
Data collection	Collect Arduino Nano 33 BLE Sense accelerometer + gyroscope data at 50 Hz using 2.56 s windows with 50% overlap	3–5 users, at least 2 environments, at least 50 labelled windows per class	Reduces domain shift between public data and deployment hardware
Phase 2 continuation	Screen more TinyML-friendly architectures under the same preprocessing and split strategy	Keep accuracy, latency, and model size jointly in view	Avoid choosing the deployable model too early
Phase 3 sensor ablation	Evaluate accelerometer-only input without gyroscope data	Compare accuracy drop against energy and sensing simplicity	Tests whether eco-mode sensing is feasible
Phase 4 deployment	INT8 quantization and TFLM / on-device inference on Arduino	Measure latency, memory, model size, and held-out Arduino accuracy	Converts the result into a true edge deployment

7.1 Track B teams must additionally address:

For Track B, Milestone 3 must additionally address the domain gap between public benchmark datasets and the final embedded deployment environment. This includes collecting Arduino-based hold-out data, comparing performance on public datasets against real-device performance, and clearly separating public-dataset reporting from Arduino deployment reporting.

If the domain gap proves significant, calibration, mixed public + Arduino retraining, and accelerometer-only ablation experiments will be considered. The final Milestone 3 evaluation should also report on-device latency, memory usage, model size, and quantized inference behavior in addition to standard classification accuracy.

8 Ethics and Limitations

This work uses public human-activity datasets and does not claim medical, safety-critical, or identity-inference capability. The main limitations are the domain gap between smartphone-waist public data and Arduino hardware, inherited preprocessing in UCI HAR, WISDM label mismatch, and likely performance changes across users, placements, clothing, and environments.

References

[1] A. N. Tarekegn, M. Sajjad, F. A. Cheikh, M. Ullah, and K. Muhammad, "Efficient human gait activity recognition based on sensor fusion and intelligent stacking framework," *IEEE Sensors Journal*, vol. 23, no. 22, pp. 28 355–28 369,

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